TRACE / FRAMEWORK NOTE

TRACE Threat Modeling Framework

A method for modern decentralized organisations that turns heterogeneous source material into a structured model, then uses STRIDE, attack trees, and optional coordination analysis to produce decision-ready security work.

Why modern organisations need TRACE

Modern cloud-first organisations combine high-value assets, distributed authority, SaaS and cloud systems, remote operations, and human control paths. TRACE keeps these in one model: STRIDE for conventional computing risk, plus attack trees and coordination analysis for protocol, system, and organisation risk.

TRACE model primitives

T — Threat actors
Potential adversaries and relevant trust assumptions.

R — Roles
Privileged authority and control paths.

A — Assets
What must be protected.

C — Critical invariants
Business, operational, economic, and technical properties that must hold.

E — Edges
Trust boundaries, value flows, interfaces, and crossings.

Inputs

- Architecture descriptions
- White papers
- System and architecture reviews
- Source code and configs
- Specs, policies, and interviews

Output

A traceable model that connects each material threat to an asset, role, invariant, trust boundary, flow, attack tree, and mitigation decision.

TRACE / OPERATING MODEL

From source material to attack paths.

AI accelerates decomposition and coverage. Senior security experts own the assumptions, severity, plausibility, and recommendations.

01 **Ingest sources**

Docs, reviews, architecture, policies, code, configs, and organisational context.

02 **Construct model**

Assets, actors, invariants, boundaries, value flows, authority paths, roles, components.

03 **STRIDE analysis**

Structured threat pass across components, flows, and privilege boundaries.

04 **Attack trees**

Top threats decomposed into plausible exploit paths and enabling conditions.

05 **Coordination inspection**

Optional pass for governance bodies, committees, vendors, administrators, operators, and other actor groups.

06 **Roadmap outputs**

Risk ranking, mitigations, open questions, and assessment, audit, or launch guidance.

Designed with human-AI co-working in mind

A deliberate sequential workflow uses AI for decomposition, coverage, and draft structure while senior security experts own the risk decisions.

- **AI proposes source inventories, model candidates, and coverage gaps.**
- **Experts approve and complement assumptions, scope, and relevance.**
- **Experts set severity, plausibility, and remediation guidance.**

Human Decision Points

Source inventory, TRACE model, STRIDE Threat identification and ranking, attack trees, roadmap.